No.	Status	Date
LNDE-RPT-20140902-024-DS-6-RA-103	Issued	2014-09-19

UT Thickness Data Sheet - 6-RA-103

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 6-RA-103 located within the ATU-100. The survey was conducted as part of the pre-change rich-amine survey to establish baseline data prior to the planned MOC, following the 3-year interval established by the DMR since the previous readings in 2011-02.

Marcus Chen, acting as the lead UT technician and SNT-TC-1A UT Level II for Larkhall NDE Services Inc., performed the measurements and is responsible for the accuracy of the recorded wall thickness data.

Technical Basis

All thickness measurements were performed under the requirements of QSCP-NDT-001 Rev 2. This procedure includes specific mandates for insulation removal, reinstatement, and the use of inspection ports to facilitate access to the pipe surface in accordance with the CUI program.

The measurement techniques and equipment calibration were executed in accordance with ASTM E797. The results are detailed in the associated report LNDE-RPT-20140902-024-R0.

Component Specifications

The subject line, 6-RA-103, is a rich amine header constructed from A106 Gr.B. The nominal dimensions are 168.3 mm OD with a nominal wall thickness of 10.97 mm.

The design limits for this circuit are 862 kPa(g) and 82 °C. The minimum allowable thickness (t-min) for the line is 4.57 mm.

Assessment Criteria

Acceptance is based on the retirement thickness, which is defined as the t-min of 4.57 mm. Any reading falling below this value is subject to an NCR.

Corrosion rates are calculated using the larger of the long-term rate or the short-term rate from the latest pair of readings at least 3 years apart. Remaining life is determined by dividing the difference between the current thickness and t-min by the calculated rate.

Inspection Interval

As a rich amine service line, 6-RA-103 is assigned a 3-year thickness interval. The next measurement date is determined by half the remaining life, not to exceed the class maximum.

Per the CUI program, insulated lines within the CUI range require profile RT or insulation windows every 5 years.

Thickness readings, measured 2014-09-19

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2011-02-18	8.32	8.07	0.024	0.070	>50	2017-09-19	
4-RA-101-02	8.56	3.81	2011-02-18	8.20	8.02	0.028	0.050	>50	2017-09-19	
4-RA-101-03	8.56	3.81	2011-02-18	7.98	7.88	0.039	0.028	>50	2017-09-19	
4-RA-101-04	8.56	3.81	2011-02-18	8.21	8.18	0.024	0.008	>50	2017-09-19	
4-RA-101-05	8.56	3.81	2011-02-18	8.16	7.98	0.033	0.050	>50	2017-09-19	
4-RA-101-06	8.56	3.81	—	—	7.91	—	—	—	2017-09-19	
4-RA-101-07	8.56	3.81	2011-02-18	8.30	8.22	0.021	0.022	>50	2017-09-19	
4-RA-101-08	8.56	3.81	2011-02-18	8.02	7.71	0.052	0.086	45.1	2017-09-19	
4-RA-101-09	8.56	3.81	2011-02-18	8.32	8.27	0.016	0.014	>50	2017-09-19	
4-RA-101-10	8.56	3.81	2011-02-18	8.28	8.21	0.025	0.020	>50	2017-09-19	
4-RA-101-11	8.56	3.81	2011-02-18	8.04	7.88	0.043	0.045	>50	2017-09-19	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2011-02-18	8.33	8.25	0.016	0.022	>50	2017-09-19	
4-RA-102-02	8.56	3.81	2011-02-18	7.99	7.89	0.037	0.028	>50	2017-09-19	
4-RA-102-03	8.56	3.81	2011-02-18	8.04	7.95	0.033	0.025	>50	2017-09-19	
4-RA-102-04	8.56	3.81	2011-02-18	7.96	7.93	0.033	0.008	>50	2017-09-19	
4-RA-102-05	8.56	3.81	2011-02-18	8.07	7.84	0.038	0.064	>50	2017-09-19	
4-RA-102-06	8.56	3.81	2011-02-18	8.01	7.95	0.030	0.017	>50	2017-09-19	
4-RA-102-07	8.56	3.81	2011-02-18	8.10	7.98	0.032	0.033	>50	2017-09-19	
4-RA-102-08	8.56	3.81	2011-02-18	7.88	7.58	0.049	0.084	45.0	2017-09-19	
4-RA-102-09	8.56	3.81	2011-02-18	7.98	7.84	0.038	0.039	>50	2017-09-19	
4-RA-102-11	8.56	3.81	2011-02-18	8.19	8.12	0.020	0.020	>50	2017-09-19	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2011-02-18	10.47	10.62	0.021	-0.042	>50	2017-09-19	
6-RA-103-02	10.97	4.57	2011-02-18	10.23	10.06	0.056	0.047	>50	2017-09-19	
6-RA-103-03	10.97	4.57	2011-02-18	10.42	10.39	0.032	0.008	>50	2017-09-19	
6-RA-103-04	10.97	4.57	2011-02-18	9.97	9.79	0.064	0.050	>50	2017-09-19	
6-RA-103-05	10.97	4.57	2011-02-18	10.10	9.81	0.062	0.081	>50	2017-09-19	
6-RA-103-06	10.97	4.57	2011-02-18	10.56	10.43	0.031	0.036	>50	2017-09-19	
6-RA-103-07	10.97	4.57	2011-02-18	10.55	10.37	0.033	0.050	>50	2017-09-19	
6-RA-103-08	10.97	4.57	2011-02-18	10.44	10.49	0.026	-0.014	>50	2017-09-19	
6-RA-103-09	10.97	4.57	2011-02-18	10.56	10.39	0.033	0.047	>50	2017-09-19	
6-RA-103-10	10.97	4.57	2011-02-18	10.05	9.93	0.059	0.033	>50	2017-09-19	
6-RA-103-11	10.97	4.57	—	—	10.10	—	—	—	2017-09-19	
6-RA-103-13	10.97	4.57	2011-02-18	10.45	10.38	0.020	0.020	>50	2017-09-19	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2011-02-18	7.95	7.84	0.036	0.031	>50	2017-09-19	
4-RA-104-02	8.56	3.81	2011-02-18	8.14	8.09	0.025	0.014	>50	2017-09-19	
4-RA-104-03	8.56	3.81	2011-02-18	8.56	8.46	0.032	0.028	>50	2017-09-19	
4-RA-104-04	8.56	3.81	2011-02-18	8.60	8.49	0.007	0.031	>50	2017-09-19	
4-RA-104-05	8.56	3.81	2011-02-18	8.57	8.37	0.015	0.056	>50	2017-09-19	
4-RA-104-06	8.56	3.81	2011-02-18	7.98	7.91	0.037	0.020	>50	2017-09-19	
4-RA-104-07	8.56	3.81	2011-02-18	7.78	7.61	0.053	0.047	>50	2017-09-19	
4-RA-104-08	8.56	3.81	2011-02-18	8.18	8.00	0.050	0.050	>50	2017-09-19	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT Level II	2014-09-19	_____